

Week 7 Notes: From ED Validation to a Real VQE Sweep

Seminar notes

June 10, 2026

1 Purpose of Week 7

Week 6 produced an ideal phase diagnostic: a non-local edge string swept across μ whose magnitude tracks the known topological interval $|\mu| < 2t$. That curve was trustworthy, but every plotted point was prepared by exact diagonalization (ED), not by a quantum circuit. The state on which the observable was measured was obtained by diagonalizing the $2^L \times 2^L$ Hamiltonian and reading off the lowest even-parity eigenvector.

Week 7 closes that gap. The milestone is:

Replace the exact-diagonalization state preparation with a parity-constrained VQE preparation at every point of the μ sweep, and validate each prepared state against ED before any hardware noise is introduced.

Everything else from Week 6 is kept fixed: the same Hamiltonian, the same edge-string observable $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$, the same shot-measurement protocol, and the same μ grid. The only change is how the state is produced. This isolates a single question: can a variational quantum circuit reproduce the ideal diagnostic curve well enough to serve as the noiseless baseline for the Block 4 noise study?

2 What Changes Conceptually from Week 6

Week 6 answered “does the chosen observable reconstruct the phase diagram?” using a trusted ED reference state. Week 7 answers a different, harder question:

“Can a gate-based, variationally optimized state reproduce that diagnostic across the whole parameter range?”

This is the step from a numerically exact target to an algorithm that must actually find the ground state. Three new difficulties appear that ED never had to confront:

1. the optimizer can fail or land in a local minimum;
2. the optimizer can drift into the wrong fermion-parity sector;
3. near criticality the ground state changes fastest, so a single ansatz must remain expressive enough everywhere.

Week 7 is largely about controlling these three failure modes and proving, point by point, that they did not occur.

3 Hamiltonian and Observable (Unchanged)

The Jordan–Wigner mapping of the open Kitaev chain gives

$$H(\mu) = -\frac{\mu}{2} \sum_{j=0}^{L-1} Z_j + \frac{t-\Delta}{2} \sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t+\Delta}{2} \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (1)$$

As in Week 6 we use $L = 4$ and the sweet-spot couplings

$$t = \Delta = 1, \quad (2)$$

so the XX term vanishes and

$$H(\mu) = -\frac{\mu}{2} \sum_j Z_j + \sum_{j=0}^{L-2} Y_j Y_{j+1}, \quad \mu_c = \pm 2t. \quad (3)$$

The plotted quantity remains the magnitude of the parity-preserving edge string

$$|\langle O_{\text{edge}} \rangle|, \quad O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}, \quad (4)$$

which for $L = 4$ is $X_0 Z_1 Z_2 X_3$. The motivation for using a non-local string rather than a local edge operator is identical to Week 6: any operator anticommuting with parity has zero expectation value in a parity eigenstate, so only a parity-preserving, non-local observable is a stable diagnostic.

4 The VQE Sweep Workflow

The core of Week 7 is a loop over the μ grid. For each μ_k the workflow is:

1. build the Pauli-term Hamiltonian $H(\mu_k)$;
2. initialize the variational parameters from the previous point, $\theta_k \leftarrow \theta_{k-1}^*$ (warm start);
3. minimize the parity-constrained cost to obtain θ_k^* ;
4. validate the resulting state against the ED reference (energy, parity, subspace fidelity, edge-string error);
5. if validation fails, recover with random restarts and deeper optimization;
6. measure $|\langle O_{\text{edge}} \rangle|$ both ideally and with finite shots;
7. store the parameters, diagnostics, and observable, then advance to μ_{k+1} .

The single most important engineering change relative to Week 6 is that the optimizer is no longer restarted independently at each μ . Neighboring ground states seed one another.

5 Warm-Start Continuation

This section unpacks the three pieces of jargon used above: the star in θ_k^* , the term “warm start,” and the phrase “neighboring ground states seed one another.”

The variational circuit and the meaning of θ

VQE does not store the quantum state directly. It stores a list of real numbers $\theta = (\theta_1, \dots, \theta_p)$, the rotation angles of the gates in the ansatz circuit $U(\theta)$. Applying that circuit to the all-zeros state produces a trial state

$$|\psi(\theta)\rangle = U(\theta)|0 \dots 0\rangle. \quad (5)$$

For the $L = 4$ `EfficientSU2` ansatz with `reps`= 3, each angle is the rotation of one R_Y gate, and θ is the full vector of those angles. Choosing θ is choosing a state; optimizing θ is searching the state space the circuit can reach.

Why θ_k carries a star

The bare symbol θ_k means “some parameter vector while we are still at point μ_k ,” including the initial guess and all the intermediate guesses the optimizer tries. The starred symbol θ_k^* is reserved for the *converged*, *optimal* parameters: the specific angles the optimizer settles on after minimizing the cost at μ_k ,

$$\theta_k^* = \arg \min_{\theta} C_{\mu_k}(\theta), \quad C_{\mu_k}(\theta) = \langle H(\mu_k) \rangle_{\theta} + \lambda(1 - \langle P \rangle_{\theta})^2. \quad (6)$$

So θ_k is a moving guess during the search, and θ_k^* is the final answer at that point. The star is the standard “optimal value” notation, exactly as in $x^* = \arg \min f(x)$. The state $|\psi(\theta_k^*)\rangle$ is the VQE approximation to the ground state of $H(\mu_k)$, and it is the state on which the edge string is measured.

What a “warm start” is

Any iterative optimizer needs a starting point. A *cold start* begins from a generic or random θ that carries no information about the answer; the optimizer must then travel a long way through parameter space to reach the minimum. A *warm start* begins from a point already believed to be close to the answer, so the optimizer only has to take a few small steps to converge.

In the μ sweep the warm start is the converged solution of the previous point:

$$\theta_k \xleftarrow{\text{initialize}} \theta_{k-1}^*. \quad (7)$$

Concretely, after we finish optimizing at μ_{k-1} and obtain θ_{k-1}^* , we do not throw those angles away. We hand them to the optimizer at μ_k as its initial guess and let it refine them. Because the optimizer starts near the answer, it needs far fewer iterations and is much less likely to wander off into a wrong local minimum.

Why “neighboring ground states seed one another”

The justification is physical, not just numerical. The ground state $|\psi(\mu)\rangle$ of the Kitaev chain changes *smoothly* with μ everywhere except at the critical points $|\mu| = 2t$. If two sweep points μ_{k-1} and μ_k are close, their ground states are nearly the same state, so the angles that prepare one are nearly the angles that prepare the other:

$$\mu_k \approx \mu_{k-1} \implies |\psi(\mu_k)\rangle \approx |\psi(\mu_{k-1})\rangle \implies \theta_k^* \approx \theta_{k-1}^*. \quad (8)$$

“Seeding” is just the act of passing θ_{k-1}^* forward as the seed (initial guess) for the next optimization. “Neighboring ground states seed one another” therefore means: because adjacent ground states are

physically close, each converged solution is an excellent initial guess for its neighbor, and the sweep can be run as one continuous chain

$$\theta_0^* \rightarrow \theta_1^* \rightarrow \theta_2^* \rightarrow \dots \quad (9)$$

rather than as a set of unrelated optimizations. This is the technique called *continuation* (or homotopy): solve an easy point first, then track the solution by taking small steps in the parameter μ . The same warm-start chain can also be run in both directions across the grid, so that each point benefits from a well-converged neighbor.

Cold vs. warm in the sweep

Independent random (cold) starts — the Week 5/6 prototype style.

- repeat the full, expensive optimization from scratch at every point;
- can converge to different local minima at adjacent μ , producing artificial jumps;
- make the swept curve noisy even before shot noise is added.

Warm-start continuation in μ .

- initialize θ_k from θ_{k-1}^* ;
- track the smooth evolution of the ground state with only a few optimizer steps per point;
- keep random restarts only as a fallback when validation fails.

What “deeper optimization” means

When a warm-started point fails validation, the recovery path does “deeper optimization.” This does *not* mean a deeper circuit; it means spending more optimization effort to dig out of a bad local minimum. In practice it combines:

- **more iterations** — raise the optimizer iteration cap (`maxiter`) so it is allowed to keep refining instead of stopping early;
- **multiple restarts** — launch several independent random initial guesses and keep the best converged result, instead of trusting the single warm-started run;
- **a tighter parity penalty / re-selection** — re-run while enforcing the even sector so the optimizer cannot escape into the wrong-parity minimum.

Only if even this fails would one consider a genuinely deeper ansatz (more **reps**); the depth scan in the next-to-last section shows **reps**=3 is already sufficient away from a handful of hard points, so “deeper optimization” is almost always about effort, not circuit depth.

Where it is hardest

The expected difficulty is concentrated near $|\mu| = 2t$, where the bulk gap closes, the ground state changes fastest, and the smoothness assumption behind warm starts is weakest. There the previous solution θ_{k-1}^* is a poorer seed, the optimizer is most likely to stall, and the recovery path (restarts plus deeper optimization) is most likely to trigger. This is exactly where the fidelity checks earn their place.

6 Parity Must Be Controlled

The Hamiltonian conserves fermion parity,

$$P = \prod_{j=0}^{L-1} Z_j, \quad [H, P] = 0. \quad (10)$$

In the topological phase the even- and odd-parity sectors are nearly degenerate. Plain energy minimization can therefore return different sectors at different μ , and a sector flip looks like a spurious discontinuity in $\langle O_{\text{edge}} \rangle$.

The Week 7 strategy is to bias the optimization toward a fixed (even) sector with a parity penalty:

$$C(\theta) = \langle H \rangle_\theta + \lambda(1 - \langle P \rangle_\theta)^2. \quad (11)$$

The penalty is minimized when $\langle P \rangle = +1$, pushing the optimizer into the even sector without distorting the energy landscape inside that sector. The parity expectation $\langle P \rangle$ is recorded at every point, and states that fall outside the parity tolerance are rejected and recovered.

7 Validation Dashboard

A low VQE energy alone is not a sufficient acceptance criterion near degeneracy. Each swept point must pass four independent checks against the ED reference:

Metric	Question	Purpose
Energy error	Is E_{VQE} near E_{ED} ?	Detect optimizer failure
Parity	Is $\langle P \rangle \approx +1$?	Keep one sector across the sweep
Subspace fidelity	Does the VQE state overlap the correct low-energy space?	Handle near-degeneracy correctly
String error	Does $\langle O_{\text{edge}} \rangle$ match the ED value?	Validate the quantity actually plotted

Subspace fidelity (rather than overlap with a single eigenvector) is the right measure because the two lowest states are nearly degenerate in the topological phase; the prepared state only needs to live in that low-energy subspace. A point that fails any check triggers deeper ansatz repetitions, more optimizer iterations, or random-restart recovery.

8 Implementation Shape

The loop body, in schematic form, is:

```

theta = initial_parameters()
for mu in mu_values:
    H = qiskit_hamiltonian(L, t, mu, delta)
    theta = optimize(parity_cost(H), initial=theta)

    diagnostics = validate(theta, H, ed_reference(mu))
    if not diagnostics.pass_all:
        theta = recover_with_restarts(H)

```

```
string_value = measure_edge_string(theta, shots=4096)
save(mu, theta, diagnostics, string_value)
```

The implementation lives in `src/block3_core.py` (helpers `vqe_cost`, `evaluate_state`, `solve_point`, `vqe_sweep`, `measure_edge_string_shots`, `best_state`, `depth_scan`) and is driven by `src/block3.py --plots 7` (the sweep) and `--plots 8` (the depth diagnostic). The ansatz is the same hardware-efficient `EfficientSU2` with R_Y rotations and linear CNOT entanglement used in Week 5. A practical point: the sweep must persist optimizer parameters and per-point diagnostics, not only the final observable, so that failed regions are debuggable and reusable as warm starts for later noise studies.

9 Reading the Week 7 Sweep Figure

The expected behavior is:

- the VQE curve (ideal and shot-based) overlays the ED baseline across the whole μ grid;
- energy error and infidelity stay near zero inside the topological region and rise mildly near $|\mu| = 2t$;
- parity deviation stays small everywhere because the penalty pins the even sector;
- recovery markers, if any, cluster near criticality where the optimization is hardest.

The top panel is the measurement-side result that Block 4 will compare against; the bottom panel is the certificate that the result was obtained from genuinely correct state preparation.

10 Ansatz Depth: How Deep Is Enough?

Scanning the number of `EfficientSU2` repetitions establishes the minimum depth needed for faithful preparation:

- one repetition (8 parameters) is too shallow; it fails worst at criticality $\mu = 2t$ (fidelity ~ 0.06);
- two or more repetitions already reach the even-parity ground space;
- the critical point is consistently the hardest, exactly where the bulk gap closes.

This justifies the choice `reps= 3` used in the full sweep: deep enough to clear criticality with margin, shallow enough to keep the optimization cheap.

11 Definition of Done for Week 7

- a scripted $L = 4$, $t = \Delta = 1$ VQE sweep over the same μ grid as Week 6;
- consistent even-parity preparation across the entire sweep;
- ED comparison for energy, parity, subspace fidelity, and edge-string expectation at every point;
- a shot-based VQE edge-string curve plotted against the ED baseline;
- optimization cost and failure/recovery statistics recorded per μ point.

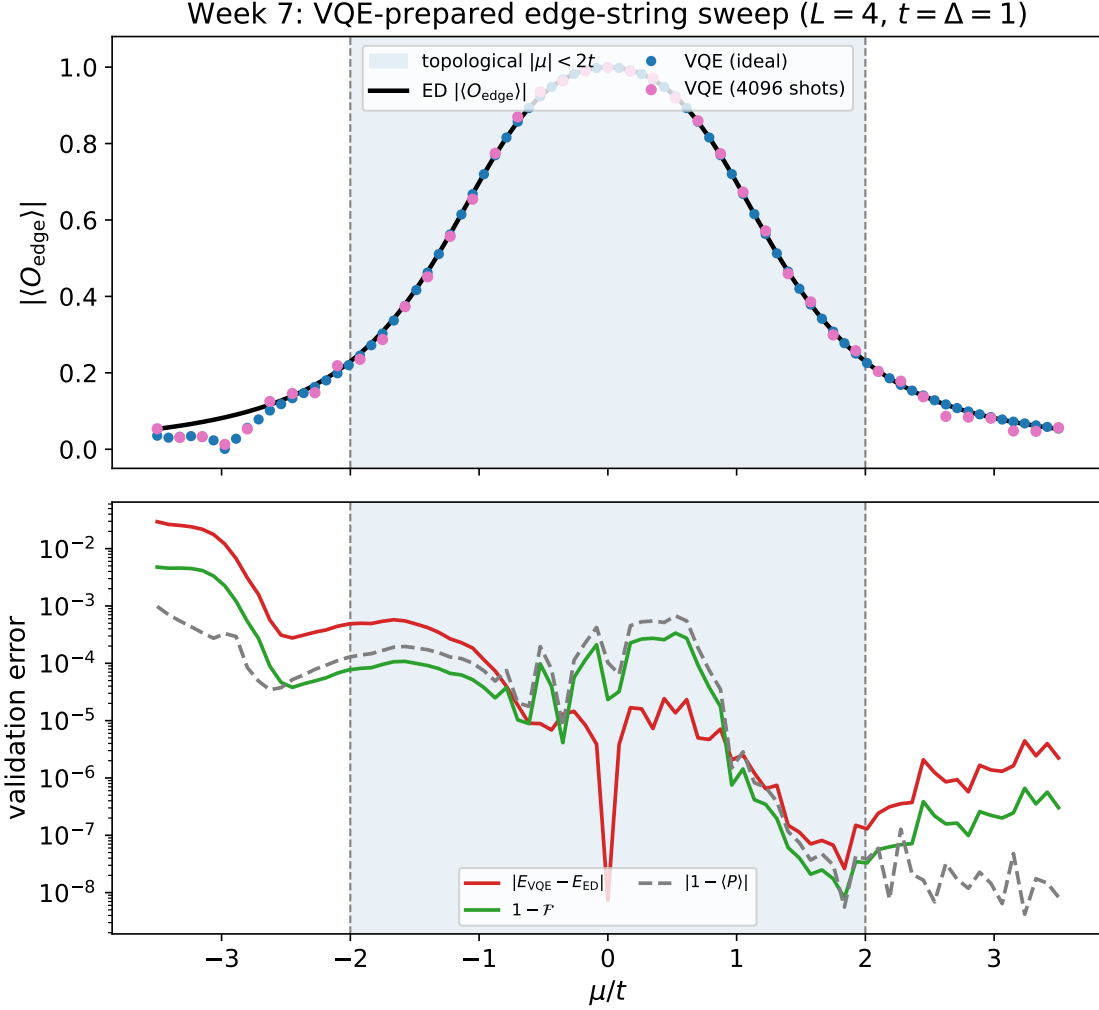


Figure 1: Week 7 VQE-prepared sweep for $L = 4$, $t = \Delta = 1$. Top: the edge string $|\langle O_{\text{edge}} \rangle|$ from parity-constrained VQE, both ideal and shot-based, plotted against the ED baseline. Bottom: per-point validation showing energy error, infidelity $1 - \mathcal{F}$, and parity deviation $|1 - \langle P \rangle|$; crosses mark points that required random-restart recovery.

12 Bridge to Block 4

Week 7 validates quantum state preparation; it does not yet test robustness. Only after the ideal VQE sweep passes validation should noise be introduced. The Block 4 plan is:

- freeze the validated ideal VQE sweep as the noiseless reference;
- apply realistic gate, readout, and relaxation (T_1 , T_2) noise to the preparation and measurement circuits;
- separate optimizer degradation from measurement degradation;
- compare raw and error-mitigated edge-string curves;
- quantify the noise level at which the inferred phase boundary becomes unreliable.

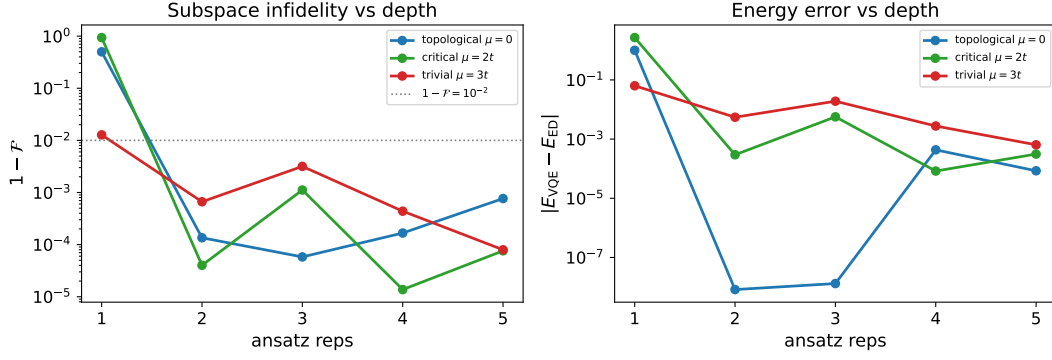


Figure 2: Ansatz-depth diagnostic: multi-start VQE subspace infidelity and energy error versus the number of ansatz repetitions, at three representative μ (topological, critical, trivial).

The conceptual progression across the block is now complete in outline: Week 6 validated the diagnostic with ED, Week 7 validates the quantum state preparation, and Block 4 tests whether the full circuit workflow survives realistic noise.